
Confidence Intervals

1. Let $X_1, \dots, X_n | \boldsymbol{\theta} \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ where $\boldsymbol{\theta} = (\mu, \sigma^2)$ unknown. Obtain a $\gamma \times 100\%$ confidence interval for σ^2 . *Be careful about inequality rules!* Is your interval exact?
2. Let $X_1, \dots, X_n | \boldsymbol{\theta} \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ where $\boldsymbol{\theta} = (\mu, \sigma^2)$ unknown. We want to construct an exact one-sided γ -coefficient confidence interval for μ of the form: $(A(\mathbf{X}), \infty)$.
 - (a) Clearly state the probability statement about $A(\mathbf{X})$ involving μ that this interval estimator should satisfy. i.e. $\Pr(\dots) = \dots$
 - (b) Now find your confidence interval!